



TRANSPORTATION

GRADE 03

MODELS FOR SUSTAINABLE

TRANSPORTATION (SDGS)

How to use this book?



This book engages students through storytelling, linking STEAM education with the Sustainable Development Goals (SDGs). It demonstrates how science, technology, engineering, art, and math can be applied to create sustainable transportation solutions, focusing on SDGs 7 and 11.

The sustainable transportation models include designs for vehicles, bicycles, and public transport systems that promote energy efficiency. The models encourage students to think critically about how transportation can contribute to achieving SDG 7 (Affordable and Clean Energy).

Additionally, the book emphasizes the role of SDG 11 (Sustainable Cities and Communities) by showcasing how well-planned public transportation networks, walkable cities, and bike paths can reduce traffic congestion and pollution. These transportation solutions ensure that urban spaces are more liveable and accessible for everyone, supporting both human health and environmental sustainability.

By promoting efficient transportation systems and clean energy, the book encourages students to think creatively about how STEAM can be applied to solve modern transportation challenges, ensuring a sustainable and healthy environment for future generations.

Theme: Transportation Construction

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Theme-01

Transportation Construction

By the end of the theme Students will be able to:

- ✓ List and define basic concepts related to forces, load, stress, and balance.
- ✓ Explain the mechanical needs of tow trucks and how they help with moving cargo.
- ✓ Compare the principles of levers and pulleys in different types of machines, like cranes.
- ✓ Describe the importance of angles and lines in designing strong structures.
- ✓ Identify the role of motors in vehicles and where they get their power.
- ✓ Describe how electromagnets work and how they are used in vehicles.
- ✓ Explain how simple programming can control the movements of an RC truck.
- ✓ Use whole numbers and geometric patterns to solve math problems related to vehicle design.

Time Frame: [75 mins]

Objectives:

Understand the concept of **transportation problems** and how they affect daily life.

Identify different **types of transportation** and the challenges they face.

Discuss how **delays in transportation** **impact store supplies and prices.**

Apply critical thinking to **real-world supply chain issues.**

Challenge Questions of the Lesson:

If you were in charge of designing a new transportation system for your city, what changes would you make to solve current challenges? Explain your reasoning with real-world examples.

SEL Relation:

Self-Awareness & Reflection: Students recognize how transportation challenges impact their daily lives and communities.

Responsible Decision-Making: Encourages students to think about real-world solutions and their consequences.

Collaboration & Communication: Group discussions and project-based activities help students develop teamwork and communication skills.

Lesson Plan

1. Engage

Teacher Role:

- Facilitate a discussion about the importance of transportation in daily life.
- Ask guiding questions to connect students' experiences with real-world scenarios.

Expected from Students:

- Participate in discussions and share their thoughts.
- Reflect on personal experiences related to transportation.

Adam was chilling on the couch with his mom, watching TV, when the news came on. The reporter said, "Oh-oh! Delays in getting stuff to stores mean shelves are running empty, and prices are rising! Experts say the way we transport things is struggling to keep up because of some problems".



Adam
"It's so frustrating! I went to the grocery store, and they were out of a bunch of things".



Adam
"Dad, I saw that too. The news said it's because of transportation problems... What kind of problems?"

Sam
"Well, it could be anything like traffic jams, too many cars on the roads, not enough truck drivers, or even trucks breaking down".

The next day at school.

Adam
"Laila, I saw on TV that transportation problems are making it tough for stores to get the stuff we need... My mom said "It could be because of traffic or not enough truck drivers".

Laila "That sounds troubling!"



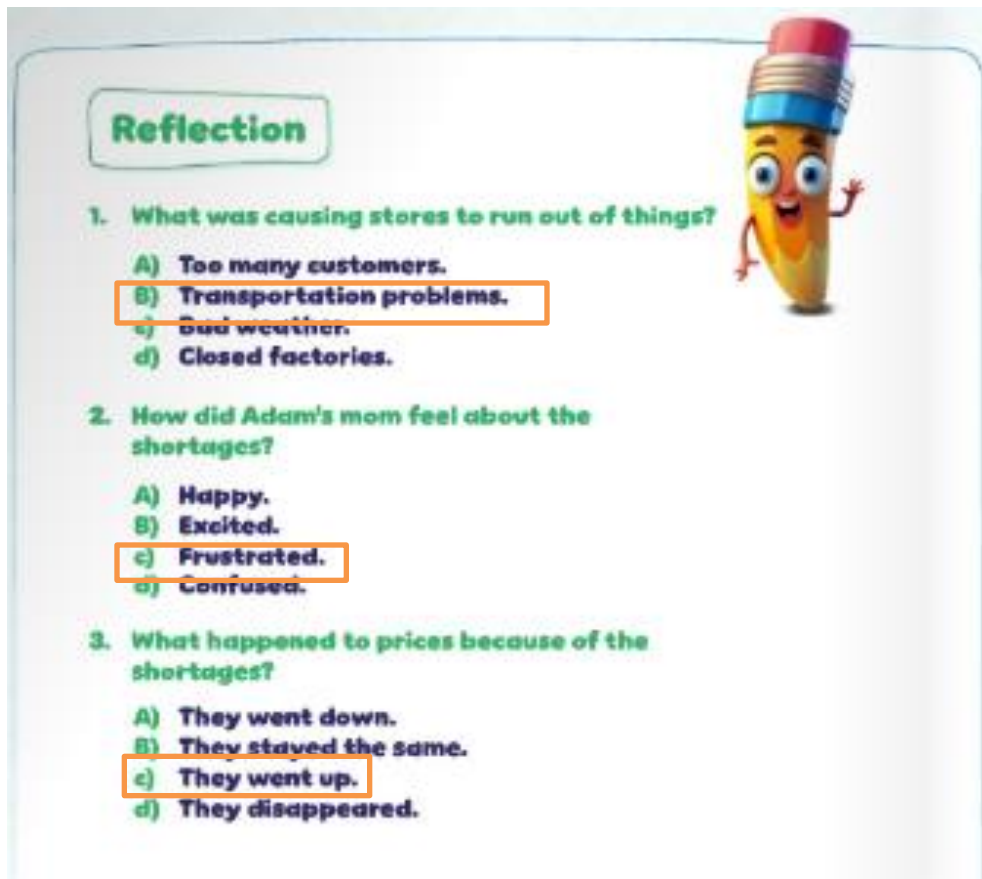
Ms. Green
"Good morning, Adam and Laila! What are you two talking about?"



Adam
"We were just talking about how transportation problems are making stores to run out of things".

Ms. Green
"That's a great topic for us to discuss today!"





- **Intro Story:** Adam and his mom watching the news about transportation delays.
 - **Story Reflection:** Discussion about supply shortages and their effects.
-

2. Think

- **Questions and Possible Answers:**

1. Why do transportation problems occur? (Traffic, lack of drivers, vehicle breakdowns)
2. How do these problems affect stores and people? (Empty shelves, rising prices)

- **Brainstorm Web:**

- Students list and categorize different transportation challenges.
-

3. Explore

- **Teacher Role:**

- Guide students in analyzing images of transportation challenges.

- **Expected from Students:**

- Observe and describe transportation issues in the images.

- Suggest possible reasons for delays.



let's Think



Why might trucks and buses have trouble getting to stores?

Explore




Watch it..

Watch the video to learn more about transportation





Scan Here!

<https://www.youtube.com/watch?v=7m7m7m7m7m>

Look at the pictures and figure out what problem the trucks and ships having with bringing things to the stores. Then, tell us what type of transportation you see!





- **Activity - Do It:**

- **Materials Needed:** Printed images or digital display.

- **Step-by-Step Instructions:**

- Observe the images of trucks, ships, and congested roads.
- Discuss the problems shown in the images.
- Write down possible causes and solutions.

- **Reflection Questions:** What do you think is the biggest transportation challenge? Why?

4. Explain

□ **Teacher Role:**

- Explain key transportation concepts.
- Facilitate discussions using examples from the book.

□ **Expected from Students:**

- Engage in discussions.
- Provide examples of transportation issues they have seen.



□ **Watch It:** [About Transshipment of Ocean Transportation. Explaining congestion of Hub Port.](https://youtu.be/8gDul1QJn17)

- **Pre-Watching:** Set a purpose (e.g., What problems do trucks and ships face?)
- **During Watching:** Encourage students to take notes.
- **Post-Watching:** Group discussion about key takeaways.

□ **Read About It:**

- **Pre-Reading:** Ask what students already know about supply shortages.
- **During Reading:** Identify new information.
- **Post-Reading:** Summarize and reflect on key points.

5. Elaborate

□ **Teacher Role:**

- Encourage students to think creatively about transportation solutions.

□ **Expected from Students:**

- Design solutions to improve transportation.

Assessment

Focus



1. What is a cargo ship used for?
A) To carry passengers across the ocean.
B) To transport goods and products.
C) To explore underwater.
D) To race in the ocean.
2. Where do cargo ships usually drop off their goods?
A) Train station.
B) Airport.
C) Port.
D) Bus stop.
3. What is the name of the large boxes used to transport goods on ships?
A) Crates.
B) Baskets.
C) Containers.
D) Bags.

Practice



Now, let's create a transportation model to show the different ways goods are transported.



- **Focus:** Answer textbook questions about transportation.
 - **Practice:** Create a transportation model.
 - **Materials Needed:** Craft materials, charts, toy vehicles.
 - **Step-by-Step Instructions:**
 1. Build a small model representing a transportation network.
 2. Identify possible bottlenecks in the system.
 3. Propose solutions to improve efficiency.
 - **Reflection Questions:** How does better transportation benefit society?
-

6. Evaluate

- **Teacher Role:**
 - Assess students' understanding through discussions and projects.
- **Expected From Students:**
 - Present their models and explain their ideas.
 - Answer assessment questions.
- **Showcase:**
 - Students display and explain their transportation models.
 - Teachers use a rubric to assess creativity, accuracy, and problem-solving skills.
- **Relation to Standards:**
 - Links to STEM education and problem-solving skills development.

Engineering Design Process

LESSON 02

Time Frame: [15 mins]

Lesson Plan

1. Engage

- **Teacher Role:**
- Introduce the engineering design process using real-life transportation examples.
- Ask students about situations where they needed to come up with solutions to problems.
- **Expected from Students:**
- Share examples of problem-solving experiences.
- Participate in discussions about design challenges in transportation.



- Intro Story: Discussion between Laila and Mrs. Sara about brainstorming solutions for transportation.
- Story Reflection: How organization helps solve transportation issues.

2. Think

- **Questions and Possible Answers:**

1. What steps do we follow when solving a problem? (Identify, brainstorm, test solutions)
2. Why is planning important before designing something? (Avoid mistakes, improve efficiency)

- **Brainstorm Web:**

- Students list transportation issues and connect them to the engineering design process.

3. Explore

- **Teacher Role:**

- Guide students in using the EDP paper to brainstorm transportation solutions.

- **Expected from Students:**

- Use the EDP template to develop ideas.
- Identify key steps in problem-solving.



- **Activity - Do It:**

- **Materials Needed:** EDP worksheets, pencils.

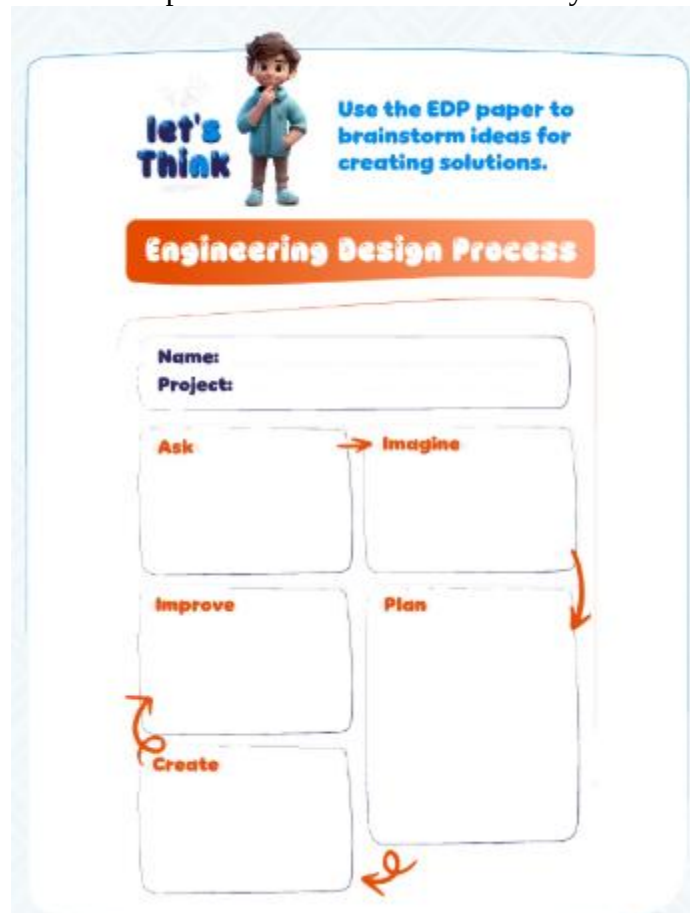
- **Step-by-Step Instructions:**

- Write a transportation challenge in the “Ask” box.
- Brainstorm solutions in the “Imagine” box.
- Plan steps to solve the problem.
- Create a model or sketch based on the plan.
- Test and improve the design.

- **Reflection Questions:** How does breaking down a problem help find better solutions?

4. Explain

- **Teacher Role:**
 - Explain each step of the EDP with real-world examples.
 - Show how engineers apply these steps in transportation.
- **Expected from Students:**
 - Discuss and provide examples of how EDP is used in daily life.



The worksheet is titled "let's Think" with a cartoon boy character. It instructs students to "Use the EDP paper to brainstorm ideas for creating solutions." The main heading is "Engineering Design Process". Below this, there are fields for "Name:" and "Project:". The process is divided into five steps in a grid: "Ask", "Imagine", "Plan", "Improve", and "Create". Arrows indicate a clockwise flow from Ask to Imagine, then to Plan, then to Improve, then to Create, and finally back to Ask.

let's Think

Use the EDP paper to brainstorm ideas for creating solutions.

Engineering Design Process

Name: _____
Project: _____

Ask	Imagine
Improve	Plan
Create	

Engineering design Process

Quickly review the phases of the Engineering Design Process, then encourage students to focus on the Ask and Brainstorming stages until they settle on a project idea. Remind them that there are no

right or wrong answers for this step—it's all about exploring possibilities!

Name: _____

Project: _____

Ask

Imagine

Mechanics Domain

LESSON 03 Wheels and axles.

Time Frame: [90 mins]

Lesson Plan

1. Engage

- **Teacher Role:**
- Introduce the concept of wheels and axles using examples like bicycles, cars, and carts.
- Discuss how these simple machines help move heavy objects with less effort.
- **Expected from Students:**
- Share experiences of using wheels (e.g., riding a bike, pulling a trolley).
- Predict how heavy objects move differently with and without wheels.

Stepping Stone:

You need to know...	You will be able to...
✓ Simple machines like ramp and pulley. ✓ The definition of force and its types such as push and pull.	✓ Identify wheels and axles as simple machine. ✓ List and define basic concepts related to forces, load, stress, and balance.



Later that day, Adam went home and told his mother about what happened at school. He explained that they had decided to make a truck and a crane with specific dimensions, and they would be able to control them remotely to avoid any delays.

Adam's Mother:
That's a great idea Adam what are you planning to do first?

Adam:
We will start with the truck.

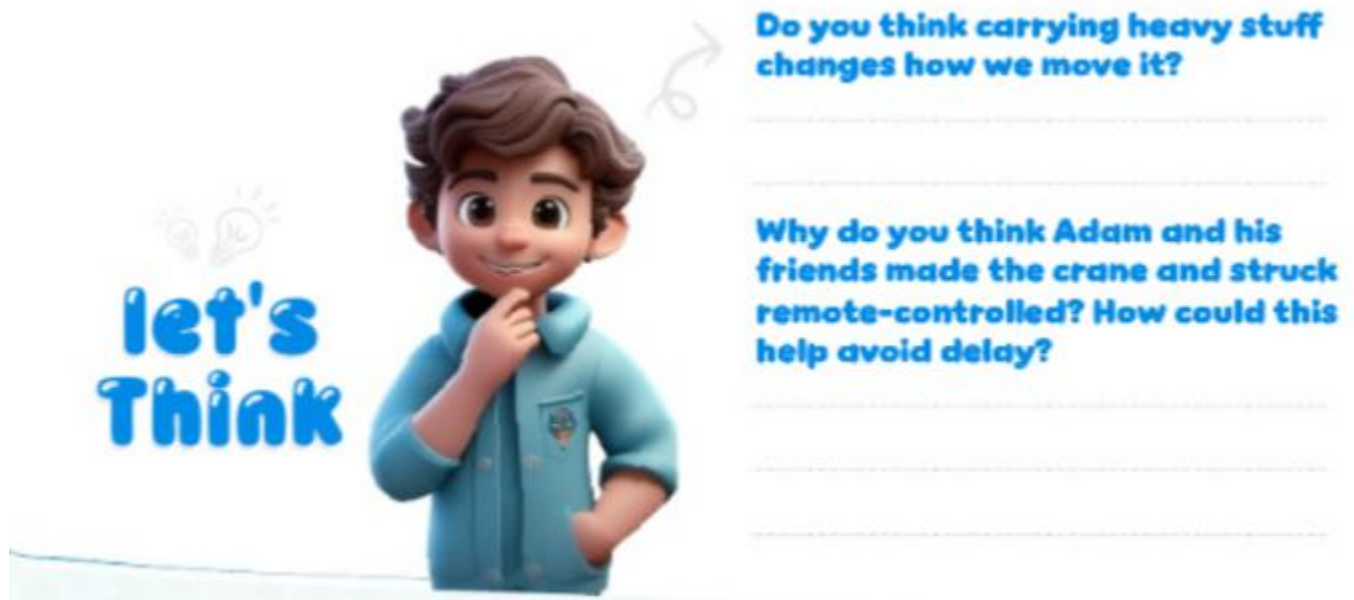
- Intro Story: Adam discusses how to make a truck and avoid delays.
- Story Reflection: How wheels and axles help in transportation.

2. Think

- **Questions and Possible Answers:**
 1. What happens when we try to move a heavy object without wheels? (It's harder to push or pull.)
 2. How do wheels make moving objects easier? (They reduce friction and distribute weight.)
- **Discussion:**
 - Why are wheels important in everyday transportation?

3. Explore

- **Teacher Role:**
 - Guide students through an experiment comparing movement with and without wheels.
- **Expected from Students:**
 - Observe and record differences in movement.
 - Identify benefits of using wheels.



- **Activity - Do It:**
- **Materials Needed:** Clothespins, small wheels, axles, objects of different weights.
- **Step-by-Step Instructions:**
- Move a clothespin without wheels and record observations.
- Attach wheels and try again.
- Compare and discuss findings.
- **Reflection Questions:** What difference did wheels make? Why?

4. Explain

- **Teacher Role:**
- Explain how wheels and axles distribute force and reduce friction.
- Show real-life examples of their applications.
- **Expected from Students:**
- Discuss how wheels are used in different machines and vehicles.

Read About It:

- **Pre-Reading:** Ask students why some objects move easily and others don't.
- **During Reading:** Identify key ideas about wheels and axles.
- **Post-Reading:** Summarize the role of simple machines in transportation.

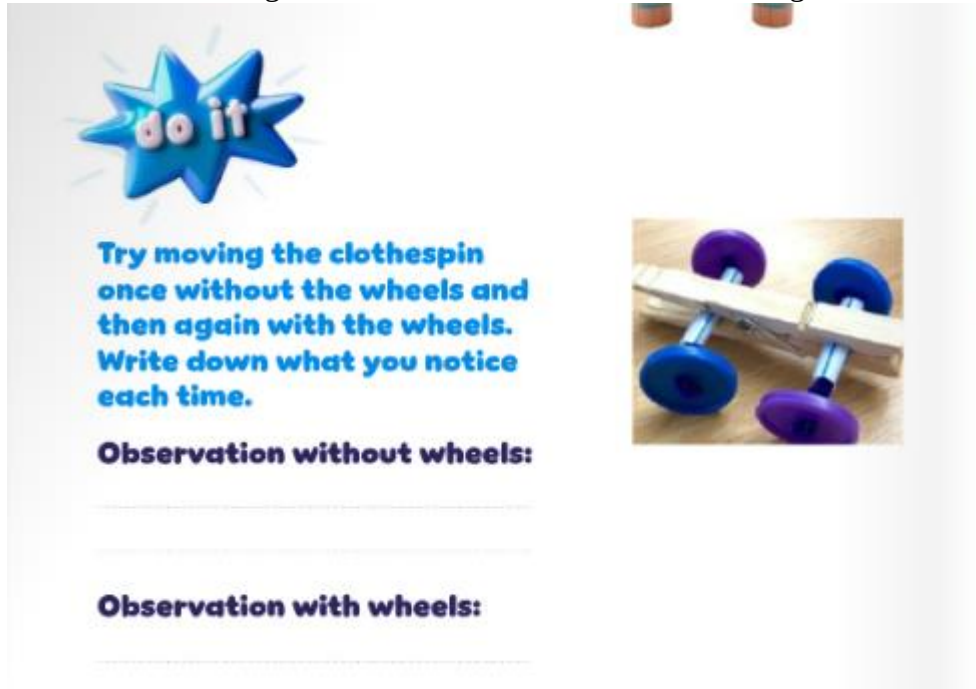
5. Elaborate

- **Teacher Role:**

6. Encourage students to apply what they learned by designing a simple wheeled vehicle.

- **Expected from Students:**

7. Use creative thinking to build a basic model of a vehicle using axles and wheels.



do it

Try moving the clothespin once without the wheels and then again with the wheels. Write down what you notice each time.

Observation without wheels:

Observation with wheels:

- **Assessment:**
- **Focus:** Answer textbook questions about wheels and axles.
- **Practice:** Design and build a DIY truck with axles and wheels.
- **Materials Needed:** Craft supplies, small wheels, axles.
- **Step-by-Step Instructions:**
 1. Design a simple vehicle.
 2. Attach axles and wheels.
 3. Test how it moves.
- **Reflection Questions:** How did adding wheels improve movement?

8. Evaluate

- **Teacher Role:**
- Assess students' understanding through projects and discussions.
- **Expected from Students:**
- Present their vehicle models and explain their design choices.
- Answer assessment questions.
- **Challenge:**
- **Test the Model:** Can it carry a load without tipping over?
- **Go Back to EDP:** Use the Engineering Design Process to refine the design.
- **Relation to Standards:**
- Links to physics principles and engineering problem-solving skills.

Assessment

Focus



Why are trucks important for transporting goods?

- A) They can fly over traffic.
- B) They are very fast.
- C) They can carry large amounts of goods over long distances.
- D) They are used to transport people.

For a medium load of several crates of fruits, which type of transportation is the best choice?

- A) Airplane.
- B) Truck.
- C) Bicycle.

Practice



Design your DIY truck and add axles and wheels to it.

Chapter 02

Design Domain

LESSON 01

Create the perfect container !

Time Frame: [90 mins]

1. Engage

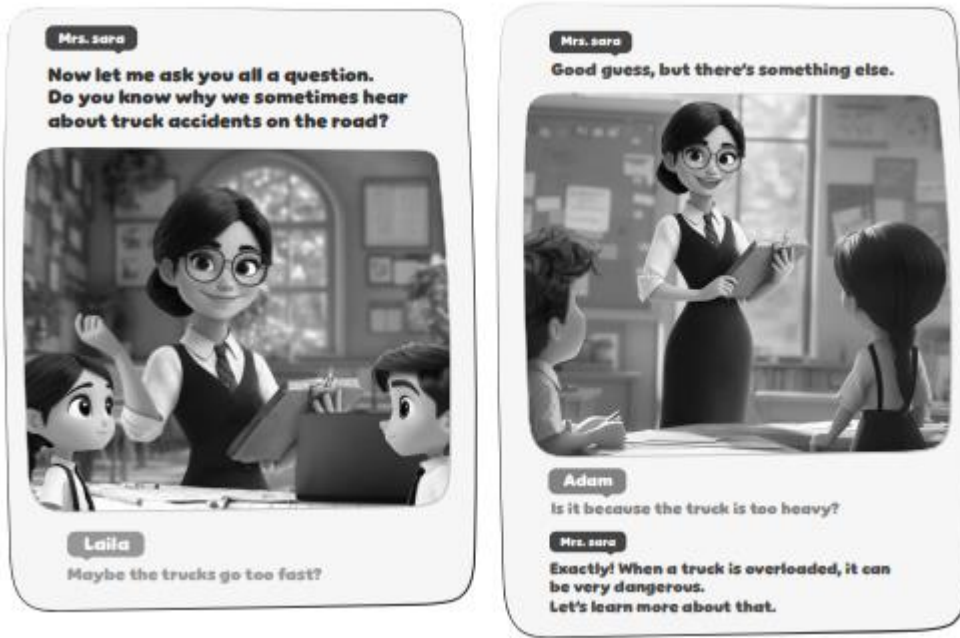
Teacher Role:

- Facilitate discussions on truck design and weight distribution.
- Ask questions to prompt critical thinking about balance and stability.
- Use storytelling and visuals to make connections to real-life truck accidents.

Expected from Students:

- Participate in discussions and share their ideas.
- Answer questions about truck accidents and weight distribution.

Parts of the Book Included:



- **Intro Story:** Discussion about truck accidents with teacher-guided questions.
- **Story Reflection:**
 - *Question:* Why do we sometimes hear about truck accidents?
 - *Suggested Answer:* Overloading and improper weight distribution affect balance.
 - *Strategy:* Use a visual storytelling approach, prompting students to analyze the images and reflect on the consequences.

2. Think

Questions and Possible Answers:

- *How does weight distribution affect truck stability?*
 - Answer: Uneven weight can cause tipping or imbalance.
- *What happens if a truck is overloaded?*
 - Answer: It becomes difficult to control, leading to accidents.

Brainstorm Web:

- Use a graphic organizer to show the relationship between weight, balance, and efficiency.
- Encourage students to map their thoughts on different materials and their properties.

3. Explore

Teacher Role:

- Guide students through hands-on activities.
- Encourage students to observe and record findings.

Expected from Students:

- Conduct experiments with toy trucks.
- Record observations in the provided **Observation Form**.

Parts of the Book Included:

Explore



You have a toy truck, some small boxes, a few rocks, and a ramp. Try these three things with the truck:

1. Place the boxes on the truck so that the weight is balanced evenly.
2. Place all the boxes on one side of the truck.
3. Place the boxes toward the back of the truck, making sure the weight is still evenly spread out.

Gently push the truck each time and watch how it moves. Write down what you notice.

- **Activity - Do It:**

- **Materials Needed:** Toy truck, small boxes, rocks, ramp.
- **Step-by-Step Instructions:**
 1. Balance the load evenly on the truck.
 2. Place all the weight on one side.
 3. Move the weight toward the back.
 4. Push the truck and record how it moves.

- **Reflection Questions:**

- What happens when the load is uneven?
- Does the truck tip over?
- How difficult is it to push?

4. Explain

Teacher Role:

- Explain balanced and unbalanced forces using visuals.
- Guide students through video content on balanced and unbalanced forces.

Expected From Students:

- Actively engage in discussions and share their observations.

Parts of the Book Included:



Balanced forces are equal in strength but push or pull in opposite directions.

When balanced forces act on an object, it stays still or keeps moving at the same speed.

Unbalanced forces are not equal, and they can make an object start moving, stop, or change direction.



- **Watch It:**
 - *Pre-Watching:* Ask students to predict how balance affects movement.
 - *During Watching:* Encourage note-taking.
 - *Post-Watching:* Discuss how forces impact stability.
- **Read About It:**
 - *Pre-Reading:* Discuss key vocabulary.
 - *During Reading:* Highlight important concepts.
 - *Post-Reading:* Summarize and reflect on applications.

5. Elaborate

Teacher Role:

- Encourage creative thinking in designing truck containers.
- Guide students in applying knowledge to real-world scenarios.

Expected From Students:

- Design truck containers using **Tinkercad**.
- Experiment with different shapes and analyze their effects.

Parts of the Book Included:



- **Assessment:**
 - *Focus:* Answer questions on balanced and unbalanced forces.
- **Practice:**
 - *Materials Needed:* Tinkercad software.
 - *Step-by-Step Instructions:* Design and modify truck containers.
 - *Reflection Questions:*
 - How does shape impact truck balance?
 - What shape is most fuel-efficient?
- **Challenge:**
 - Change the container's shape and analyze its impact on stability and fuel efficiency.

6. Evaluate

Teacher Role:

- Assess student work through discussions and projects.
- Provide feedback based on observations and results.

Expected From Students:

- Answer assessment questions.
- Demonstrate understanding through their designs and observations.

Parts of the Book Included:

Assessment

Focus



Which of these is an example of an unbalanced force?

- A) A book sitting on a table.
- B) A car moving at constant speed on a straight road.
- C) A person pushing a heavy box that doesn't move.
- D) A child kicking a ball.

What can unbalanced forces do to an object?

- A) Make it start moving.
- B) Make it slow down.
- C) Make it change direction.
- D) All of the above.

If you push a toy car with the same force on both sides, what will happen?

- A) The car will move forward.
- B) The car will move backward.
- C) The car will stay still.
- D) The car will spin around.

- **Showcase:** Students present their container designs.
- **Rubric:**
 - *Creativity in Design:* How innovative is their container?
 - *Application of Concepts:* Does it reflect balance and efficiency?
 - *Presentation & Explanation:* Can students justify their design choices?

Time Frame: [90 mins]

Lesson Plan:

1. Engage

Teacher Role:

- Facilitate discussion by asking students how vehicles move.
- Encourage students to brainstorm different ways to make a toy truck move.
- Introduce the concept of using electricity to power a motorized vehicle.

Expected from Students:

- Participate in discussions and share their ideas.
- Answer questions about different ways to move a truck.
- Think critically about how a toy truck can move independently.

Parts of the Book Included:



- **Intro Story Dialogue** (Teacher and students discussing movement options)

Intro Story Strategies:

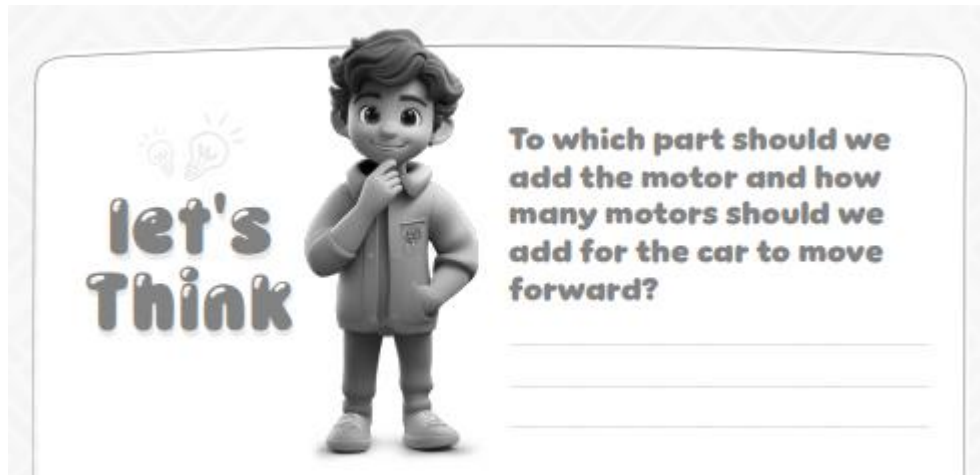
- **Think-Pair-Share:** Students discuss in pairs how vehicles move and then share with the class.
- **Visual Demonstration:** Show an image or real example of vehicles moving.
- **Questioning:** “What makes a real truck move?”

Story Reflection:

- **Questions:**
 - What did Laila suggest to move the truck?
 - How did the teacher challenge her idea?
 - What was Adam’s suggestion?
- **Suggested Answers:**
 - Laila suggested pushing it.
 - The teacher asked how it could move on its own like a real truck.
 - Adam suggested using batteries and motors.
- **Strategies:**
 - **Guided Discussion:** Ask leading questions to drive student thinking.

- **Role-Playing:** Have students act out the dialogue.
-

2. Think



Questions and Possible Answers:

- **To which part should we add the motor?** (To the wheels or the body where it connects to the wheels.)
- **How many motors should we add?** (Two, for balanced movement.)

Brainstorm Web:

- Central Idea: **Moving a Truck**
 - **Methods:** Hands, Wind, Electricity
 - **Components:** Batteries, Motors, Circuits

Strategies:

- **Concept Mapping:** Draw a web diagram connecting different movement methods.
 - **Collaborative Discussion:** Small groups discuss and write ideas on sticky notes.
-

3. Explore

Teacher Role:

- Guide students in identifying parts of the truck where motors should be placed.
- Demonstrate connecting a motor to a battery and smart pad.

Expected From Students:

- Participate in hands-on activity by assembling motors onto trucks.
- Observe and document what happens when electricity is applied.

Parts of the Book Included:



Activity - Do It:

- **Materials Needed:**
 - Toy truck
 - 2 small motors
 - Battery pack
 - Smart pad
 - Connecting wires
- **Step-by-Step Instructions:**
 1. Identify the part of the truck where motors should be placed.
 2. Connect the motors to the smart pad.
 3. Attach the battery pack.
 4. Observe how the truck moves.
- **Reflection Questions:**
 - What happens when the battery is connected?
 - How does the motor help the truck move?

4. Explain

Teacher Role:

- Explain energy transformation from chemical (battery) to electrical to mechanical (motor movement).
- Facilitate discussion about real-life applications of electric motors.

Expected From Students:

- Share observations from the activity.
- Discuss how energy is converted in a motorized system.

Parts of the Book Included:

- **Watch It:** <https://www.youtube.com/watch?v=PKm4ZVNmJyQ>

Watch It:

- **Pre-Watching:** Ask, “What do you think happens inside a motor?”
- **During Watching:** Take notes on energy transformation.
- **Post-Watching:** Discuss observations.

Read About It:

- **Pre-Reading:** Introduce key vocabulary (motor, battery, circuit, energy transformation).
- **During Reading:** Highlight important concepts.
- **Post-Reading:** Summarize how electricity powers movement.

5. Elaborate

Teacher Role:

- Encourage students to experiment with different ways of wiring motors.
- Introduce challenges (e.g., “How can we make it move faster?”).

Expected from Students:

- Modify their trucks to improve movement.
- Test different wiring configurations.

Assessment

Focus



When you turn on a motor, what happens?

- A) It turns off.
- B) It makes a sound but doesn't move.
- C) It starts to spin or move.
- D) It turns into a toy.

What happens to energy when a motor is working?

- A) It disappears.
- B) It changes into heat and movement.
- C) It gets smaller.
- D) It becomes a sound.

What kind of energy does a battery in a toy motor use?

- A) Solar energy.
- B) Chemical energy.
- C) Sound energy.
- D) Wind energy.

Assessment:

- **Focus Questions:**

- What happens when you turn on a motor? (Answer: It spins or moves.)
- What happens to energy when a motor works? (Answer: It changes into heat and movement.)
- What kind of energy does a battery use? (Answer: Chemical energy.)

Practice:

Practice



Now try to add to motors to your circuit.

After we learned how can we connect the motors, let's go back to the EDP process and add it to the create part .



Robot
Setup

1. Add
Motor

2. Add
Battery

3. Add
Wiring

Showcase

Add two motors and battery to your truck and make it move forward.

- **Materials Needed:**
 - Extra motors
 - Additional battery packs
 - Alternative wiring setups
- **Step-by-Step Instructions:**
 1. Experiment with different wiring setups.
 2. Observe and compare movements.
 3. Record results.
- **Reflection Questions:**
 - How did changing the wiring affect movement?
 - What setup worked best?

Challenge:

- Students take their toy trucks home and modify them further.

6. Evaluate

Teacher Role:

- Assess students' understanding through observation and discussion.
- Provide feedback on their truck modifications.

Expected from Students:

- Demonstrate understanding by explaining their projects.
- Answer assessment questions accurately.

Parts of the Book Included:

- **Showcase Project Evaluation**

Showcase:

- Students present their modified trucks and explain their design choices.
-

Relation to Standards:

- Aligns with **STEM Education Standards** on energy transformation.
- Supports **Next Generation Science Standards (NGSS)**:
 - **PS3.B:** Conservation of Energy and Energy Transfer.
 - **ETS1.A:** Defining and Delimiting an Engineering Problem.